Heritability and selection of strategic traits in the NFL coaching tree

Jon Williamson[1]

[1]Williamson Consulting Group. Correspondence: jonwilliamson@live.com

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Social media summary. NFL coaching lineages inherit a mentor's system, not his success: what transmits is largely not what wins.

Abstract

Human expertise is transmitted socially, and where learners have identifiable teachers the transmission leaves a genealogy. We treat the National Football League coaching tree as a cultural pedigree and, for a preregistered set of ten standardized play-calling traits, estimate each trait's transmissibility (h^2, mentor-to-protege resemblance from a Bayesian parent-offspring model with a measurement-error correction) and its selection (S, association with a coach's wins above replacement). Every trait is a genuine, repeatable individual characteristic (repeatabilities 0.34 to 0.58), yet transmissibility and selection come apart. Schematic-identity traits transmit but do not win: shotgun $h^2 = 0.45$ (95% HDI $[0.27, 0.65]$) with $S = 0.08$, and tempo similarly. Offensive aggression is the mirror image, repeatable (0.40) and selected ($S = 0.18$, $[0.08, 0.27]$) yet not transmitted ($h^2 = -0.15$); a preregistered contrast confirms it is repeatable but not heritable (0.55, $[0.24, 0.87]$). Defensive aggression alone both transmits ($h^2 = 0.49$) and wins ($S = 0.15$). The predicted negative heritability-fitness relationship does not appear (Spearman $+0.32$, $[-0.54, 0.69]$): rather than a trade-off, transmission and selection are largely decoupled. Coaching lineages propagate systems, not success, and the fitness-linked behavior spreads through a channel a purely vertical lens would miss.

Keywords: cultural evolution; social learning; vertical transmission; heritability; cultural fitness; sport.

Preregistration. The confirmatory analyses (C1–C3, hypotheses H1–H3) and their decision rules were registered before any confirmatory quantity was computed on the final measurement pipeline: osf.io/y2kr5 (`doi:10.17605/OSF.IO/Y2KR5`). We report the preregistered verdicts as they fell, supported or not, and flag any deviation where it occurs.

Data and code availability. All data and code are public and seed-fixed: github.com/jwilliamson7/Coach_Tree and github.com/jwilliamson7/Coach_WAR (the source of the WAR phenotype, pinned at commit `dffe2f1`). The full pipeline runs from one command (`scripts/analysis/run_ehs_confirmatory.py`).

1 Introduction

Human skilled practice is transmitted socially, and where practitioners learn from identifiable teachers the transmission leaves a genealogy. Coaching in the National Football League (NFL) is an unusually legible case. Head coaches hire assistants, assistants become coordinators, coordinators are promoted to head coach, and the record of who worked under whom is documented season by season. The result is a "coaching tree": a directed network of mentor–protege ties along which styles of play are widely believed to descend. Popular accounts speak of the Walsh tree or the Belichick tree much as a breeder speaks of a bloodline, and NFL hiring leans heavily on that pedigree logic, on the assumption that working under a successful coach transfers both winning approaches and strategic expertise.

Recent work has begun to treat sport tactics as a cultural-evolutionary system. Mesoudi (2020) analyzed the cultural evolution of association-football formations and showed that managers' choices spread through strategic social learning, while explicitly naming transmission along mentor–protege lineages as open work. That gap motivates the present study. Cultural transmission theory distinguishes *vertical* transmission, from a parent-like model to a learner, from *horizontal* transmission, the spread of behavior among contemporaries (Cavalli-Sforza and Feldman, 1981; Boyd and Richerson, 1985). A coaching tree is a natural substrate for measuring the vertical channel directly, because both the mentor's and the protege's behavior are observed on the field.

We borrow the two central quantities of quantitative genetics and apply them to a cultural system. The first is transmissibility, the cultural analog of heritability h^2: how strongly a protege's own head-coaching style resembles the style of the coach they apprenticed under. The second is selection S: how strongly a trait covaries with success, measured here as a coach's context-adjusted wins above replacement (WAR). In biological populations these two quantities are famously in tension. Traits most closely tied to fitness tend to have *lower* heritability, because directional selection erodes additive genetic variance, a pattern documented across many species and formalized as a negative cross-trait relationship between heritability and fitness (Mousseau and Roff, 1987; Kruuk et al., 2000). Whether an analogous tension arises in a cultural transmission system, where the inheritance is social learning rather than meiosis, is unknown.

We ask a single organizing question: in the NFL coaching tree, are the strategic traits that transmit the same traits that win? We separate strategy into two families. *Approach-to-scenarios* traits describe how aggressively a coach attacks game situations (going for it on fourth down, passing in neutral situations, throwing beyond the first-down marker, attempting two-point conversions, and on defense loading the box, sending extra rushers, and playing man coverage). *Schematic-identity* traits describe the formational signature of a team (how much it operates from shotgun, and how fast it plays). Our preregistered hypotheses were:

- H1. Approach traits are more strongly *selected* (predict WAR), while identity traits are more strongly *inherited* (transmit down the tree).
- H2. Offensive aggression is a repeatable individual trait that is nonetheless not heritable: its within-coach repeatability clearly exceeds its transmissibility.
- H3. Across the ten sub-traits, heritability and selection are negatively related: the traits that transmit most strongly are not the ones that predict winning.

These predictions were registered before any confirmatory quantity was computed on the final measurement pipeline (Section 2). As reported below, the data support a decoupling of transmission and selection but not the specific negative trade-off of H3, and the picture is enriched by one trait, defensive aggression, that both transmits and wins.

Table 1: The ten frozen sub-traits and their four gene-level composites. The two families are approach-to-scenarios (aggression) and schematic identity.

Composite (family)	Sub-traits	Data window
Offensive aggression (approach)	fourth-down, pass-heavy, deep-pass, two-point	2006–2024
Tempo (identity)	no-huddle, pace	2006–2024
Defensive aggression (approach)	box-stacking, pass-rush	2016–2024
	man-coverage	2018–2024
Shotgun (identity)	shotgun	2006–2024

2 Materials and methods

2.1 Coaching phenotypes ("genes")

We construct standardized coach-season phenotypes from play-by-play data for the 2006–2024 NFL seasons (obtained through `nfl_data_py`; more than 600,000 plays). For each strategic decision we train a play-level expectation model (gradient-boosted trees) on pre-play game context only, deliberately excluding season, week, and coach identity, so the baseline reflects what an average coach would do in identical circumstances and is time-invariant. A coach-season phenotype is the average deviation of observed behavior from this expectation, mean(actual) − mean(predicted), standardized (z-scored) across the coach-season panel; positive values denote more aggressive, faster, or more shotgun-oriented play than context predicts. This residual construction isolates a strategic tendency from forced circumstance: a coach who goes for it often only because he is always trailing is not thereby aggressive.

Every play is attributed to the head coach who led the team for that game, so a within-season head-coaching change splits the season between the two coaches at the game of the change. Coordinators are not assigned their own phenotype; the phenotype is defined only at the head-coach level, so a play is attributed to one coach and never to both a head coach and a coordinator at once. To keep a coach's own tendencies out of the yardstick that measures them, expectation models are fit with the scoring coach held out and with cross-fitting folds balanced across eras, so each play is scored by a model that never trained on that coach.

Ten sub-traits are frozen for the primary analysis (Table 1), grouped into four gene-level composites (offensive aggression, tempo, defensive aggression, and shotgun) that serve as labeled exemplars but are not counted as independent points in the cross-trait test.

2.2 Contemporary-group (era) adjustment

Each phenotype is expressed as a deviation from its own season's league mean (within-season demeaning) before any modeling. Because both a mentor's and a protege's values are re-centered on their own seasons, league-wide drift is removed exactly once and is not counted as transmission or selection. This is the single era control, the cultural analog of the contemporary-group (herd-year-season) adjustment used to estimate breeding values in the animal model; no further season or era term enters the models. Coach WAR is already a within-season, replacement-relative quantity (about 3% between-season variance) and receives no further adjustment.

2.3 Measurement error and reliability

Each coach-season phenotype is an estimate with a known sampling variance, computed per sub-trait from the play-level variance divided by the squared play count and rescaled to the standardized scale. This per-observation variance is carried explicitly in the heritability (C1) and repeatability (C2) models, so both are estimated for the latent trait rather than its noisy observation. Composite genes propagate their components' sampling variance; component reliabilities are estimated with a DerSimonian–Laird random-effects decomposition (DerSimonian and Laird, 1986).

2.4 C1: transmissibility (h^2)

A coordinator-to-head-coach transition pairs two different coaches. The outcome is the protege's own phenotype once they became a head coach; the predictor is the phenotype of the head coach they apprenticed under, measured on that mentor's team over the seasons the protege served there as a coordinator. Each coach contributes at most one transition per coordinator role (the most recent coordinator stint preceding a head-coaching stint), which avoids pseudo-replication. Because the phenotype is head-coach-level, this design measures whether the head-coaching style a coach apprenticed under predicts that coach's own later style; it is a between-person parent-offspring comparison, not a coordinator's own trait.

We estimate h^2 with a Bayesian multilevel parent-offspring model fit at the coach-season grain (Wilson et al., 2010; Nakagawa and Schielzeth, 2010). For protege i, a latent mentor level m_i and a latent own level a_i are linked by

$$a_i = \alpha + \beta\, m_i + \varepsilon_i, \qquad \varepsilon_i \sim \mathrm{Normal}(0, \sigma_a), \tag{1}$$

and each individual mentor-overlap season x_{ij} and protege head-coaching season y_{ik} loads on the corresponding latent level with a variance that combines a shared true within-coach season-to-season component τ^2 and the known measurement variance of that observation:

$$x_{ij} \sim \mathrm{Normal}\big(m_i,\ \sqrt{\tau^2 + \mathrm{se}_{ij}^2}\big), \qquad y_{ik} \sim \mathrm{Normal}\big(a_i,\ \sqrt{\tau^2 + \mathrm{se}_{ik}^2}\big). \tag{2}$$

Modeling individual seasons rather than pre-averaging lets the mentor predictor enter as a latent variable carrying its measurement variance, an errors-in-variables correction for the attenuation that predictor noise would otherwise cause. The transmissibility is the parent-offspring regression slope at the coach-season grain,

$$h^2 \;=\; \frac{\beta\, \sigma_m^2}{\sigma_m^2 + \tau^2} \;=\; \beta_{\text{latent}} \times \text{repeatability}, \tag{3}$$

which is bounded above by the mentor repeatability $\sigma_m^2/(\sigma_m^2 + \tau^2)$: a trait cannot transmit more faithfully than it is a stable trait to begin with. Because a mentor transmits a whole style rather than half a genome, the slope itself is the transmissibility, with no Mendelian doubling. Priors are $\mathrm{Normal}(0, 1)$ on the intercept and slope (phenotypes are z-scored) and half-$\mathrm{Normal}(0, 1)$ on the variance components; sampling uses four chains with 2000 post-warmup draws each, a non-centered parameterization, and target diagnostics $\hat{R} \leq 1.01$ and bulk effective sample size > 400 (met for every fit). As robustness checks the slope is refit with a mentor varying intercept and, in a frequentist version, with mentor-clustered standard errors.

2.5 C2: repeatability

Per-trait repeatability is a Bayesian variance-components intraclass correlation on the full head-coach coach-season panel, with the per-season measurement variance carried as an explicit level so the ratio is the latent-trait repeatability, $\sigma^2_{\text{between}}/(\sigma^2_{\text{between}} + \sigma^2_{\text{within}})$. It is the ceiling for h^2; because heritability cannot exceed repeatability, we report $P(h^2 > \text{repeatability})$ per trait, a value near one flagging a specification problem rather than a substantive result.

2.6 Selection (S)

Selection S is the era-adjusted, inverse-variance-weighted correlation of a trait with Coach WAR: within-season demean the standardized gene and WAR, weight each coach-season by its games-based WAR precision, and take the coach-clustered weighted correlation. Single-season WAR is noisy, and because mean-zero noise in the outcome attenuates a correlation rather than inflating it, this estimate is a conservative floor. WAR is drawn from the sibling Coach_WAR repository at commit `dffe2f1` (Williamson, 2026).

2.7 C3 and decision rules

For the ten sub-traits we assemble the pair (h^2, S) and compute their Spearman rank correlation with a 95% bootstrap interval that resamples the traits and draws h^2 from its posterior and S from its bootstrap. The rank form is the established test of the heritability-fitness relationship in quantitative genetics (Mousseau and Roff, 1987; Kruuk et al., 2000). Each hypothesis is judged by one pre-specified interval rule. H1 is supported if the approach family's mean S exceeds the identity family's and the identity family's mean h^2 exceeds the approach family's, each difference with a 95% interval excluding zero. H2 is supported if (aggression repeatability minus aggression h^2) has a 95% interval excluding zero. H3 is supported if the Spearman correlation is negative with a 95% bootstrap interval excluding zero.

2.8 Exploratory analyses

Two analyses are declared exploratory and corrected among themselves under a Benjamini–Hochberg false-discovery rate (Benjamini and Hochberg, 1995), separately from the confirmatory tests. The first is a within-era Moran's I on the mentor network (an undirected, row-normalized adjacency of mentor–protege ties, with a permutation null that shuffles trait labels only within career-midpoint era blocks) (Moran, 1950). The second is reproductive fitness as an exposure-normalized rate: head-coaching offspring (proteges who themselves became head coaches) per head-coaching season, with a minimum-exposure floor, together with the test of whether coach quality predicts that rate net of exposure.

2.9 Software

Analyses are in Python; the Bayesian models use PyMC. The pipeline is seed-fixed and runs from a single command.

3 Results

3.1 The phenotypes are repeatable coach traits

Before asking how a trait transmits or whether it wins, we confirm it is a stable individual characteristic rather than a team artifact. Every trait is repeatable: the latent-trait repeatabilities (Table 2) range from 0.34 (pass-heavy) to 0.58 (fourth-down), all clearly above zero, so each phenotype carries substantial stable between-coach signal net of measurement noise. This is the quantitative-genetics prerequisite for a meaningful heritability: a trait must be repeatable before it can be transmissible.

Three companion analyses corroborate that the genes belong to the coach. A one-way variance decomposition attributes far more of each gene's variance to coach identity than to team identity (for composite offensive aggression, 0.50 versus 0.14; against a cardinality-matched permutation null the coach excess is about 0.29 and the team excess about 0.08, so the coach signal is roughly three times larger). When coaches change teams their tendencies travel with them (composite offensive aggression across-team $r = 0.55$, 95% CI $[0.29, 0.73]$, on the era-adjusted gene; in a mover regression the coach's prior-team gene carries $\beta = 0.54$, $[0.26, 0.88]$, alongside the destination's pre-arrival identity). A coach's aggression also persists year to year ($r = 0.50$ between consecutive seasons). And career-average gene leaders match public reputation (the most aggressive coaches by career composite are Brandon Staley and Doug Pederson; tempo is led by Chip Kelly and Kliff Kingsbury). Together with the repeatabilities, these establish the phenotype.

3.2 Transmissibility and selection come apart

Table 2 reports, for each trait, the preregistered transmissibility h^2 (C1), repeatability (C2), and selection S. Two facts stand out. First, transmissibility varies sharply across traits that are all similarly repeatable: the schematic-identity genes transmit strongly (shotgun $h^2 = 0.45$, 95% HDI $[0.27, 0.65]$; no-huddle 0.30, $[0.09, 0.51]$) as does defensive aggression (0.49, $[0.13, 0.85]$), while offensive aggression does not transmit at all (-0.15, $[-0.46, 0.12]$). Second, selection follows a nearly opposite ordering: the approach components predict WAR (offensive aggression $S = 0.18$, $[0.08, 0.27]$; defensive aggression 0.15, $[0.03, 0.27]$; pass-heavy 0.16; pass-rush 0.17) while the identity genes do not (shotgun 0.08, $[-0.03, 0.17]$; tempo 0.03). The frequentist mentor-clustered slopes and the mentor varying-intercept refits agree with the Bayesian h^2 (for example shotgun clustered slope 0.47, $p < 0.001$; defensive 0.61, $p = 0.03$; offensive aggression -0.03, $p = 0.85$).

3.3 The heritability by selection map and the preregistered verdicts

Placing the four composites in $h^2 \times S$ space (Figure 1) makes the structure visible. Offensive aggression sits in the adaptive-but-personal corner (selected, not heritable); shotgun and tempo sit in the heritable-but-neutral corner (transmitted, not selected); defensive aggression sits alone in the adaptive-and-heritable corner. The two axes are close to orthogonal.

The three preregistered decision rules resolve as follows.

H2 is supported. Offensive aggression is repeatable (0.40) but not heritable ($h^2 = -0.15$); the preregistered contrast, repeatability minus h^2, is 0.55 with a 95% HDI of $[0.24, 0.87]$, excluding zero. Offensive aggression is a real, stable, personal trait that does not pass down the tree.

H1 is not supported as a conjunction, though its selection half holds. The approach family is more selected than the identity family (mean-S difference $+0.11$, 95% interval $[0.01, 0.22]$,

Table 2: Transmissibility (h^2, C1), repeatability (C2), and selection (S) for the four composites and the ten sub-traits. h^2 and repeatability are posterior medians with 95% HDIs; S is the era-adjusted inverse-variance-weighted correlation with WAR with a 95% coach bootstrap CI. All C1/C2 fits met $\hat{R} \leq 1.01$ and bulk ESS > 400.

Trait	h^2	repeatability	S
Offensive aggression	-0.15 $[-0.46, 0.12]$	0.40 $[0.28, 0.52]$	$+0.18$ $[0.08, 0.27]$
fourth-down	-0.02 $[-0.45, 0.37]$	0.58 $[0.42, 0.74]$	$+0.04$ $[-0.05, 0.13]$
pass-heavy	$+0.08$ $[-0.08, 0.25]$	0.34 $[0.25, 0.43]$	$+0.16$ $[0.04, 0.27]$
deep-pass	$+0.32$ $[0.02, 0.62]$	0.42 $[0.30, 0.54]$	$+0.09$ $[0.00, 0.17]$
two-point	$+0.00$ $[-0.23, 0.28]$	0.35 $[0.23, 0.48]$	-0.02 $[-0.11, 0.07]$
Tempo	$+0.19$ $[-0.02, 0.41]$	0.45 $[0.36, 0.55]$	$+0.03$ $[-0.07, 0.14]$
no-huddle	$+0.30$ $[0.09, 0.51]$	0.47 $[0.37, 0.55]$	-0.02 $[-0.11, 0.08]$
pace	$+0.10$ $[-0.15, 0.34]$	0.37 $[0.26, 0.47]$	$+0.07$ $[-0.04, 0.19]$
Defensive aggression	$+0.49$ $[0.13, 0.85]$	0.37 $[0.24, 0.51]$	$+0.15$ $[0.03, 0.27]$
box-stacking	$+0.44$ $[0.05, 0.81]$	0.39 $[0.26, 0.54]$	$+0.06$ $[-0.04, 0.16]$
pass-rush	$+0.42$ $[0.12, 0.73]$	0.39 $[0.25, 0.52]$	$+0.17$ $[0.04, 0.29]$
man-coverage	$+0.14$ $[-0.33, 0.62]$	0.52 $[0.39, 0.65]$	$+0.13$ $[-0.01, 0.28]$
Shotgun	$+0.45$ $[0.27, 0.65]$	0.43 $[0.34, 0.52]$	$+0.08$ $[-0.03, 0.17]$

excluding zero), but the identity family is not reliably more heritable than the approach family (mean-h^2 difference $+0.16$, $[-0.12, 0.43]$, including zero). The heritability half fails for a substantive reason: defensive aggression is an approach trait that is nonetheless highly heritable ($h^2 = 0.49$), so the approach family does not have uniformly low transmissibility.

H3 is not supported. Across the ten sub-traits the Spearman correlation between h^2 and S is $+0.32$ with a 95% bootstrap interval of $[-0.54, 0.69]$: the point estimate is positive, the opposite of the predicted negative trade-off, and the interval spans zero. The biological heritability-fitness trade-off does not replicate here.

Taken together, the verdicts describe a *decoupling* rather than a trade-off. Transmission and selection are close to independent across coaching traits, with two clean archetypes, inherited-but-neutral identity (shotgun, tempo) and selected-but-personal offensive aggression, and one trait, defensive aggression, that occupies the both corner and is chiefly responsible for the absence of a clean negative relationship.

3.4 Exploratory: network signal and reproductive success

On the mentor network, within-era Moran's I shows positive phylogenetic signal for the schematic and defensive genes and none for offensive aggression (defensive $I = 0.21$, permutation $p = 0.002$; tempo 0.12, $p = 0.008$; shotgun 0.08, $p = 0.04$; offensive aggression -0.01, $p = 0.52$). Under the exploratory Benjamini–Hochberg correction the defensive and tempo signals survive. Reproductive success (distinct head-coaching proteges) is heavy-tailed (Gini 0.66; the top decile of mentors produced 43% of all head-coaching descendants; Bill Belichick and Andy Reid lead). It tracks career WAR ($r = 0.36$), but the link runs through longevity, not per-season quality: offspring count correlates 0.71 with seasons coached, and net of tenure the exposure-normalized rate is essentially unrelated to quality (partial $r = 0.09$). Coaches shape the league less by winning in any given season than by lasting long enough to send many descendants into head-coaching chairs, carrying

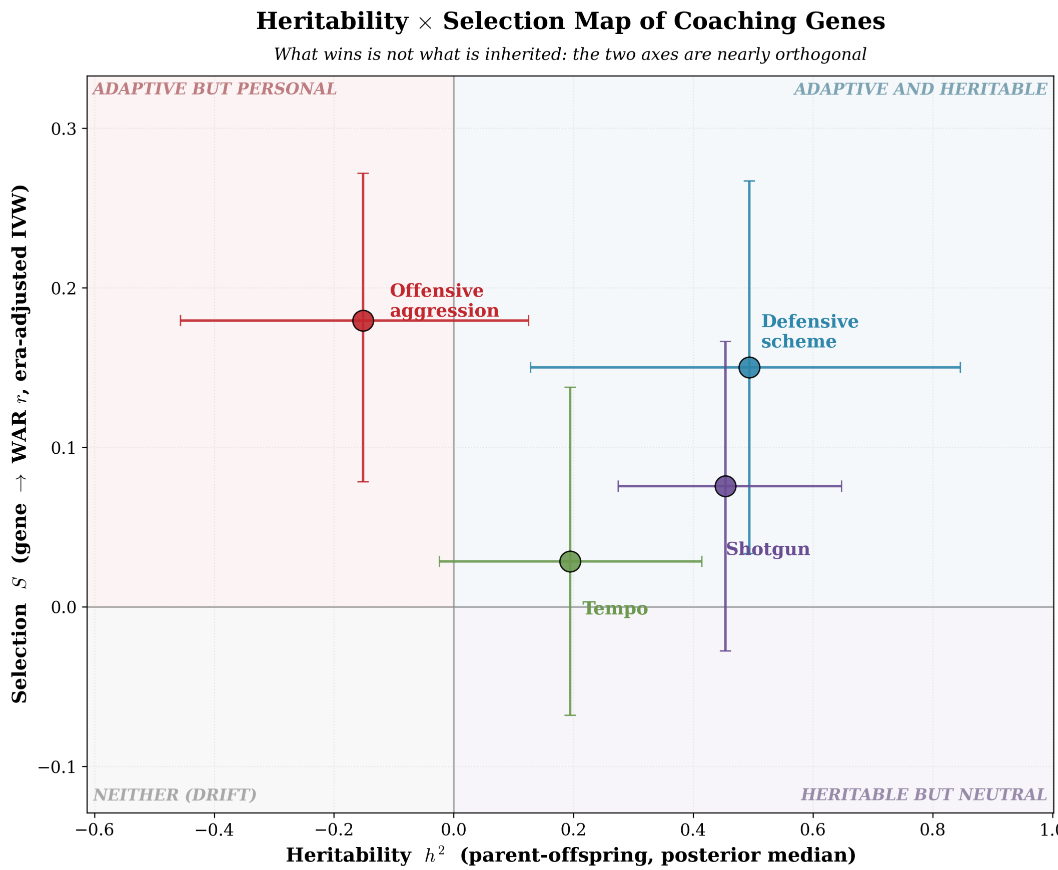

Figure 1: Heritability (h^2, C1 posterior median) versus selection (S, era-adjusted inverse-variance-weighted gene-WAR correlation) for the four gene-level composites, with 95% intervals on both axes. Offensive aggression is adaptive but personal; shotgun and tempo are heritable but neutral; defensive aggression is both. The axes are nearly independent.

their schematic identity with them.

4 Discussion

4.1 Decoupling, not a trade-off

We set out to test whether the traits that transmit down a coaching lineage are the traits that win, expecting the negative heritability-fitness relationship familiar from biological populations. That specific trade-off did not appear. Instead, transmission and selection are largely *decoupled*: across ten strategic traits the two axes are close to orthogonal (cross-trait Spearman +0.32, interval spanning zero), not negatively related. The result is not the absence of structure but a different structure. Two archetypes are clean. Schematic-identity traits, shotgun and tempo, are transmitted faithfully (h^2 up to 0.45) yet carry no detectable fitness consequence ($S \approx 0$): in evolutionary terms they behave like heritable but selectively neutral markers (Kimura, 1968), tags of lineage that ride along with apprenticeship rather than with winning. Offensive aggression is the mirror image, a repeatable and selected but non-transmitted trait (h^2 indistinguishable from zero, $S = 0.18$): what wins on offense is a personal characteristic that does not pass down. This last pattern is the preregistered H2, and it is the sharpest single result: a trait can be a stable individual property (repeatability 0.40) and predict success and still not be inherited.

4.2 Why the trade-off is absent: two channels of transmission

The classical heritability-fitness trade-off assumes a single inheritance channel along which selection erodes heritable variance. Two features of the coaching case break that assumption. First, the fitness-linked trait, offensive aggression, has essentially *no* vertical transmission, so there is no heritable variance for lineage selection to erode; the trade-off cannot arise where the adaptive trait is not inherited. Yet offensive aggression nonetheless swept the league over our window, its mean rising and its spread across coaches widening. The resolution is that coaching traits move through two channels, not one (Cavalli-Sforza and Feldman, 1981; Boyd and Richerson, 1985). Schematic identity spreads *vertically*, down the offensive-coordinator apprenticeship. Offensive aggression, lacking a vertical channel, spread *horizontally*, coaches copying a winning tactic from their contemporaries rather than their mentors. A purely vertical, coaching-tree lens would have missed that diffusion entirely, and the era structure we treat as a confounder for inference is, read substantively, the horizontal-transmission signal. Second, defensive aggression occupies the adaptive-and-heritable corner that a strict trade-off forbids, and it is largely why the cross-trait relationship comes out flat-to-positive rather than negative.

4.3 Coaching lineages transmit systems, not success

The pedigree logic that drives NFL hiring finds little support for its strongest claim. Working under a successful mentor does not, in these data, predict a protege's own success, and the in-game risk tendency that does predict success is not passed down. What coaching trees reliably transmit is schematic system: formation and, to a lesser degree, tempo identity, and specifically through the coordinator who would actually install it. Organizations can far more confidently predict the *system* a coordinator will bring than whether he will win, and the system itself is not the edge. This reframes the coaching tree as a conduit for heritable-but-neutral cultural markers, legible precisely because they track apprenticeship rather than performance, with defensive scheme the one domain where what is inherited also helps win.

4.4 Limitations

The transmission samples are small (about 40 offensive and 12–17 defensive coordinator-to-head-coach transitions), so the h^2 posteriors are wide and the cross-trait test in particular has low power; H3's wide interval should be read as inconclusive about a weak relationship, not as evidence for a positive one. Coaching pedigrees are non-tree, with multiple simultaneous mentors and overlapping careers, and coaching relatedness can only be assumed by analogy rather than derived from a Mendelian process, so we use a parent-offspring regression rather than a pedigree animal model. The genes attribute team play-calling to the head coach and do not isolate a coordinator's individual contribution; the transmission tests therefore ask whether the head-coaching style a coach apprenticed under predicts his own later style, a mentor-protege resemblance that is consistent with social learning but cannot on its own be separated from assortative hiring, in which head coaches recruit coordinators who already share their approach. We therefore read h^2 as vertical transmissibility in the broad sense. The defensive-aggression measures begin in 2016 and man coverage in 2018, and the defensive-aggression to WAR link may partly reflect reverse causation (better defensive talent enables man coverage and extra rushers). Finally, single-season WAR is noisy;

we address this with inverse-variance weighting and treat the season-level estimates as conservative floors.

4.5 Conclusion

Treating the NFL coaching tree as a cultural pedigree, we find that the strategic traits that transmit are largely not the traits that win. Schematic identity is inherited but neutral; offensive aggression is selected but personal and, strikingly, not heritable despite being a stable individual trait; defensive aggression alone both transmits and wins. The negative heritability-fitness relationship of biological populations does not appear; transmission and selection are instead decoupled, and the fitness-linked behavior spreads horizontally rather than down the lineage. Coaching trees, in the end, propagate systems, not success.

Acknowledgments

This work uses play-by-play data from `nfl_data_py` and coaching histories from Pro Football Reference; the WAR phenotype is from the companion Coach_WAR project.

References

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.

Boyd, R., & Richerson, P. J. (1985). *Culture and the Evolutionary Process*. University of Chicago Press.

Cavalli-Sforza, L. L., & Feldman, M. W. (1981). *Cultural Transmission and Evolution: A Quantitative Approach*. Princeton University Press.

DerSimonian, R., & Laird, N. (1986). Meta-analysis in clinical trials. *Controlled Clinical Trials*, 7(3), 177–188.

Falconer, D. S., & Mackay, T. F. C. (1996). *Introduction to Quantitative Genetics* (4th ed.). Longman.

Kimura, M. (1968). Evolutionary rate at the molecular level. *Nature*, 217(5129), 624–626.

Kruuk, L. E. B., Clutton-Brock, T. H., Slate, J., Pemberton, J. M., Brotherstone, S., & Guinness, F. E. (2000). Heritability of fitness in a wild mammal population. *Proceedings of the National Academy of Sciences*, 97(2), 698–703.

Mesoudi, A. (2020). Cultural evolution of football tactics: strategic social learning in managers' choice of formation. *Evolutionary Human Sciences*, 2, e25.

Moran, P. A. P. (1950). Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2), 17–23.

Mousseau, T. A., & Roff, D. A. (1987). Natural selection and the heritability of fitness components. *Heredity*, 59(2), 181–197.

Nakagawa, S., & Schielzeth, H. (2010). Repeatability for Gaussian and non-Gaussian data. *Biological Reviews*, 85(4), 935–956.

Price, G. R. (1970). Selection and covariance. *Nature*, 227(5257), 520–521.

Williamson, J. (2026). Coach WAR: a counterfactual approach to evaluating NFL head coach impact in the modern era. Companion manuscript.

Wilson, A. J., Reale, D., Clements, M. N., Morrissey, M. M., Postma, E., Walling, C. A., Kruuk, L. E. B., & Nussey, D. H. (2010). An ecologist's guide to the animal model. *Journal of Animal Ecology*, 79(1), 13–26.